

The SynthOmnicon - A Holographic Type Theory

Operations & Algebra Reference Sheet

The 12 Primitives

Sym	Name	Values
D	Dimensionality	$D_{\wedge}, D_{\Delta}, D_{\infty}, D_{\text{holo}}$
T	Topology	$T_{\text{net}}, T_{\text{in}}, T_{\boxtimes}, T_{\text{box}}, T_{\text{holo}}$
R	Recognition	$R_{\text{super}}, R_{\text{cat}}, R_{\dagger}, R_{lr}$
P	Polarity	$P_{\text{asym}}, P_{\psi}, P_{\pm}, P_{\text{sym}}, P_{\pm}^{\text{sym}}$
Γ	Int. Grammar	$\Gamma_{\text{and}}, \Gamma_{\text{or}}, \Gamma_{\text{seq}}, \Gamma_{\text{broad}}$
S	Stoichiometry	$1:1, n:n, n:m$
F	Fidelity	$F_{\ell} < F_{\text{eth}} < F_h$
K	Kinetic Char.	$K_{\text{fast}} < K_{\text{mod}} < K_{\text{slow}} < K_{\text{trap}}$
G	Granularity	$G_{\sqsubset} < G_{\sqsupset} < G_{\mathbb{N}}$
H	Chirality	$H_0 < H_1 < H_2 < H_{\infty}$
Ω	Top. Protection	$\Omega_0 < \Omega_{Z_2} < \Omega_Z$
Φ	Criticality	$\Phi_{\text{sub}} < \Phi_c < \Phi_c^{\text{cx}} < \Phi_{\text{EP}} < \Phi_{\text{super}}$

Classes: **categorical** (6) — **ordered** (5) — **special** (Φ_c absorbing)

Distance Metric (§26.1)

Diagonal weighted Euclidean:

$$d(A, B) = \sqrt{\sum_{i=1}^{12} w_i (\xi_i^A - \xi_i^B)^2}$$

$\mathbf{w} = (1.0^{\times 9}, w_H=0.8, 1.0, w_{\Omega}=0.7)$

Riemannian (§26.2): $d_g = \sqrt{(\xi^A - \xi^B)^{\top} \Sigma^{-1} (\xi^A - \xi^B)}$

Directed: $d^{\rightarrow}(A \rightarrow B)_j = w_j \cdot \max(0, \xi_j^A - \xi_j^B)$ (upward free)

Core Operations

Meet $\sqcap$ (GLB)

Categorical: $A_i = B_i \rightarrow A_i$, else **CONFLICT**

Ordered: $\min(A_i, B_i)$

$\Phi_c \sqcap x = \Phi_c$

Join $\sqcup$ (LUB)

Categorical: $A_i = B_i \rightarrow A_i$, else **CONFLICT**

Ordered: $\max(A_i, B_i)$

Tensor $\otimes$ = Join with precondition:

$$A \otimes B = \begin{cases} A \sqcup B & \text{if no cat. conflicts} \\ \text{undefined} & \text{otherwise} \end{cases}$$

Physical: co-assembly requires categorical compatibility.

Algebraic Properties

- **Commutative:** $A \star B = B \star A$ for $\star \in \{\sqcap, \sqcup, \otimes\}$
- **Associative:** $(A \star B) \star C = A \star (B \star C)$ (when def.)
- **Idempotent:** $A \star A = A$
- Φ_c **meet-absorption:** $\Phi_c \sqcap x = \Phi_c$ for all x
- **Distance:** $d(A \otimes B, A)^2 \leq d(A, B)^2$
- **Path-dep.:** $(A \otimes B) \otimes C$ may exist when $B \otimes C$ undefined

Cross-Primitive Axioms (Machine-Checked)

- A1.** $H_{\infty} \Rightarrow K \leq K_{\text{trap}}$ (irrev. $\rightarrow$ trap)
A2. $\Omega \geq \Omega_Z \Rightarrow H \geq H_2$ (strong top. $\rightarrow$ chirality)
A3. $D_{\text{holo}} \Leftrightarrow T_{\text{holo}}$ (holo dim. $\leftrightarrow$ top.)
A4. $\Omega_Z \Rightarrow D_{\text{holo}}$ ($\mathbb{Z}$ -protection $\rightarrow$ holo)

Path-Dependence Theorem

$(A \otimes B) \otimes C$ defined even when $B \otimes C$ undefined, iff $\exists$ ordered primitive i :

- $C_i > B_i$ (C needs more than B)
- $A_i = C_i$ (A supplies what C needs)
- No categorical conflict between $(A \otimes B)$ and C

Interpretation: Catalysis, scope promotion, chirality templating unified as “context A enables $B \rightarrow C$ ”.

Mechanism	Blocking	A provides
Catalysis	Φ_c required	$A_{\Phi} = \Phi_c$
Scope promotion	$G_{\mathbb{N}}$ required	$A_G = G_{\mathbb{N}}$
Chirality templating	H_2 required	$A_H = H_{\infty}$

Primitive Semantics (Quick Ref.)

Categorical: exact match or **CONFLICT** **Ordered:** min/max resolution Φ_c : absorbing element

Structural Universality

Synthon: $\langle D; T; R; P; F; K; G; \Gamma; \Phi; H; S; \Omega \rangle$

Validity: Must satisfy axioms A1–A4.

Thesis: $d(\mathbf{x}, \mathbf{y}) = 0 \Rightarrow$ constraint-identical, substrate-independent.

Conflict Distance: $d_c(S) = \sqrt{|\text{Conf}(S)|}$ measures encoding reliability.

Dual Encoding:

- Holistic $\mathcal{E}_H(S)$: top-down from behavior
- Compositional $\mathcal{E}_C(S)$: bottom-up from $\otimes S_i$
- $\text{Conf}(S) = \{p : \mathcal{E}_H(S)[p] \neq \mathcal{E}_C(S)[p]\}$

Tensor Product Properties

When $A \otimes B$ defined:

- $(A \otimes B)_F = \max(F_A, F_B)$
- $(A \otimes B)_K = \max(K_A, K_B)$
- $(A \otimes B)_j = \max(A_j, B_j)$ for all ordered j
- Categorical: must agree (precondition)
- Φ_c absorption: if $A_{\Phi} = \Phi_c$ or $B_{\Phi} = \Phi_c$, then $(A \otimes B)_{\Phi} = \Phi_c$

Translation Cost: $\delta_A = d(A, A \otimes B)$, $\delta_B = d(B, A \otimes B)$; smaller δ is dominant.

Ouroboricity

Scalar: $\mathcal{O}(\mathbf{x}) = [\Phi=\Phi_c] \cdot (1 + [\Omega \neq \Omega_0] + [H \geq H_1] + [G=G_{\mathbb{N}}])$

Tiers (priority R1>R5):

- **R1** $\Phi_c + P_{\pm}^{\text{sym}} \rightarrow O_{\infty}$ (special Frobenius)
- **R2** $\Phi \in \{\Phi_{\text{sub}}, \Phi_{\text{super}}, \Phi_{\text{EP}}\} \rightarrow O_0$
- **R3** $\Phi_c + \Omega_0 \rightarrow O_1$
- **R4** $\Phi_c + \Omega \neq \Omega_0 + D \in \{D_{\wedge}, D_{\text{holo}}, D_{\Delta}\} \rightarrow O_2$
- **R5** $\Phi_c + \Omega \neq \Omega_0 + D_{\infty} \rightarrow O_2^{\dagger}$

Note: $\mathcal{O}$ cannot detect O_{∞} (no P term); O_{∞} absorbed by Φ_{EP} under $\otimes$.